Line Balancing Study - Assembly Line 2 - Section 1: Executive Summary & Scope

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 1 (Executive Summary & Scope) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Defines the operational charter, scope boundaries, and executive accountability.

Operational requirement 1.1: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck. Secondary control 1.2: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.

- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Section 1 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored

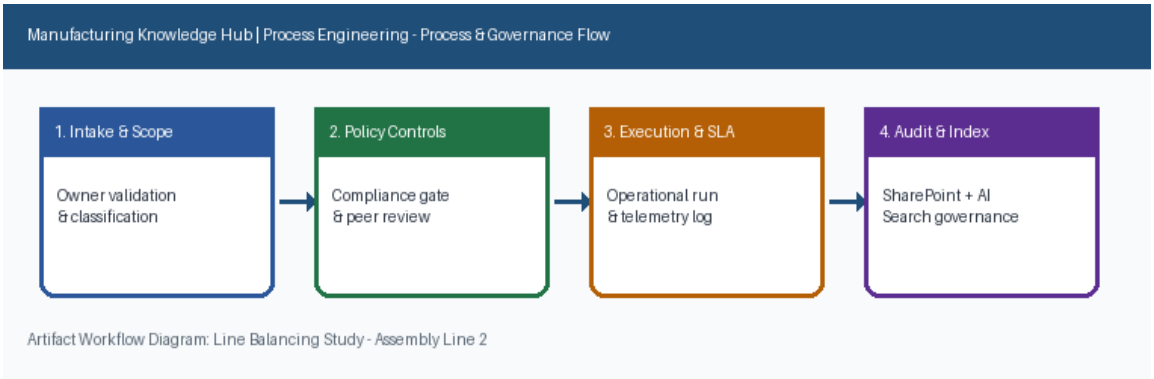
Line Balancing Study - Assembly Line 2 - Section 2: Architecture & Process Flow

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 2 (Architecture & Process Flow) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Maps end-to-end process stages, handoff gates, and system-of-record integrations.

Operational requirement 2.1: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt. Secondary control 2.2: Projected throughput improvement is approximately 6% without added headcount.

- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Projected throughput improvement is approximately 6% without added headcount.
- Section 2 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-2	Carlos Ibarra	3 business days	Stage 2 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Line Balancing Study - Assembly Line 2 - Section 3: Governance Standards & Controls

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 3 (Governance Standards & Controls) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Establishes mandatory compliance controls, approval thresholds, and audit evidence.

Operational requirement 3.1: Projected throughput improvement is approximately 6% without added headcount. Secondary control 3.2: Recommendation is scheduled for pilot validation next production cycle.

- Projected throughput improvement is approximately 6% without added headcount.
- Recommendation is scheduled for pilot validation next production cycle.
- Section 3 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-3	Carlos Ibarra	10 business days	Stage 3 Verified

Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

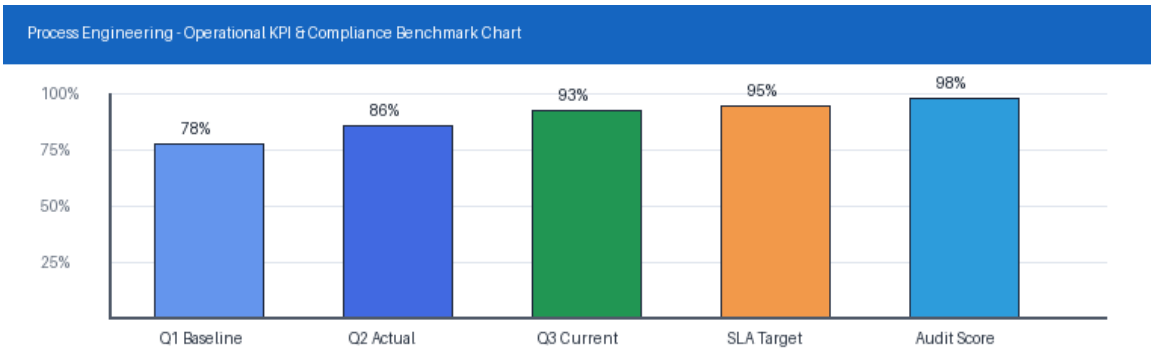
Line Balancing Study - Assembly Line 2 - Section 4: Operational KPIs & Benchmarks

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 4 (Operational KPIs & Benchmarks) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Tracks baseline performance, target SLAs, and quarterly benchmark progression.

Operational requirement 4.1: Recommendation is scheduled for pilot validation next production cycle. Secondary control 4.2: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.

- Recommendation is scheduled for pilot validation next production cycle.
- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Section 4 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-4	Carlos Ibarra	5 business days	Stage 4 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Line Balancing Study - Assembly Line 2 - Section 5: Roles, Responsibilities & RACI

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 5 (Roles, Responsibilities & RACI) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Clarifies ownership across accountable executives, operational leads, and reviewers.

Operational requirement 5.1: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck. Secondary control 5.2: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.

- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Section 5 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-5	Carlos Ibarra	3 business days	Stage 5 Verified

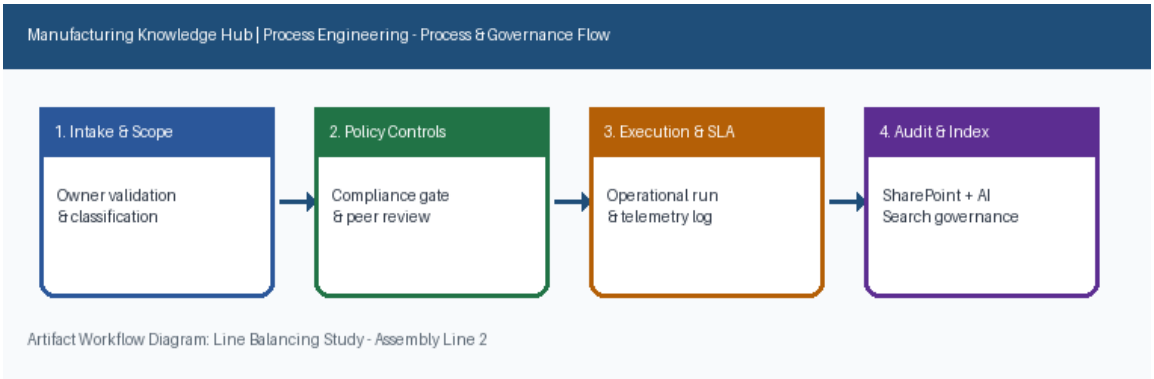
Line Balancing Study - Assembly Line 2 - Section 6: Standard Operating Procedures

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 6 (Standard Operating Procedures) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Provides step-by-step execution checklists, intake criteria, and verification gates.

Operational requirement 6.1: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt. Secondary control 6.2: Projected throughput improvement is approximately 6% without added headcount.

- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Projected throughput improvement is approximately 6% without added headcount.
- Section 6 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-6	Carlos Ibarra	10 business days	Stage 6 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Line Balancing Study - Assembly Line 2 - Section 7: Risk Register & Mitigations

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 7 (Risk Register & Mitigations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Catalogs operational, compliance, and delivery risks alongside preventive mitigations.

Operational requirement 7.1: Projected throughput improvement is approximately 6% without added headcount. Secondary control 7.2: Recommendation is scheduled for pilot validation next production cycle.

- Projected throughput improvement is approximately 6% without added headcount.
- Recommendation is scheduled for pilot validation next production cycle.
- Section 7 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-7	Carlos Ibarra	5 business days	Stage 7 Verified

Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

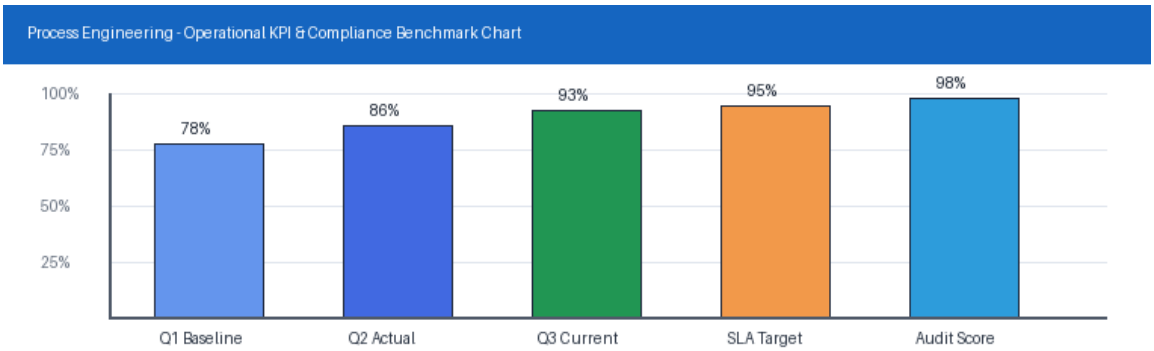
Line Balancing Study - Assembly Line 2 - Section 8: Quality Assurance & Audits

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 8 (Quality Assurance & Audits) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Specifies sampling frequencies, inspection tolerances, and nonconformance workflows.

Operational requirement 8.1: Recommendation is scheduled for pilot validation next production cycle. Secondary control 8.2: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.

- Recommendation is scheduled for pilot validation next production cycle.
- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Section 8 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-8	Carlos Ibarra	3 business days	Stage 8 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

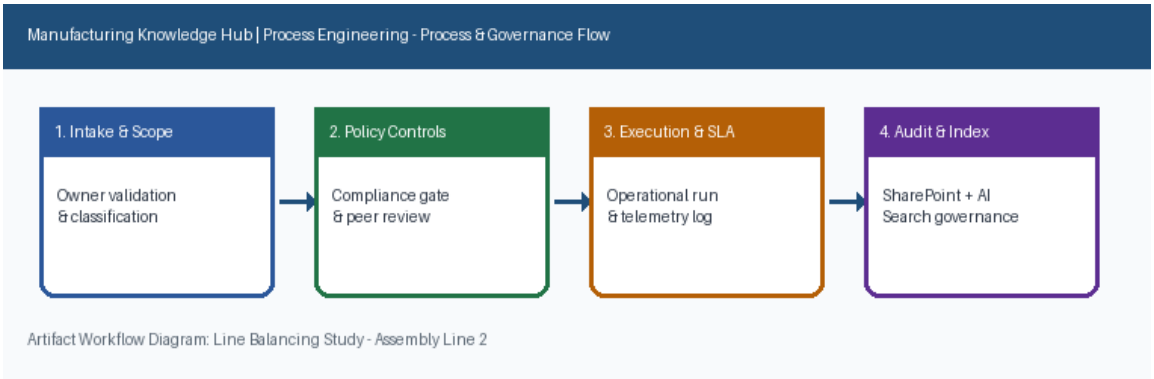
Line Balancing Study - Assembly Line 2 - Section 9: Exception Handling & Escalations

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 9 (Exception Handling & Escalations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Outlines severity tiers, response SLAs, and cross-functional escalation paths.

Operational requirement 9.1: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck. Secondary control 9.2: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.

- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Section 9 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-9	Carlos Ibarra	10 business days	Stage 9 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Line Balancing Study - Assembly Line 2 - Section 10: Cross-Functional Dependencies

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 10 (Cross-Functional Dependencies) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Documents upstream and downstream team dependencies and service-level commitments.

Operational requirement 10.1: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt. Secondary control 10.2: Projected throughput improvement is approximately 6% without added headcount.

- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Projected throughput improvement is approximately 6% without added headcount.
- Section 10 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-10	Carlos Ibarra	5 business days	Stage 10 Verified

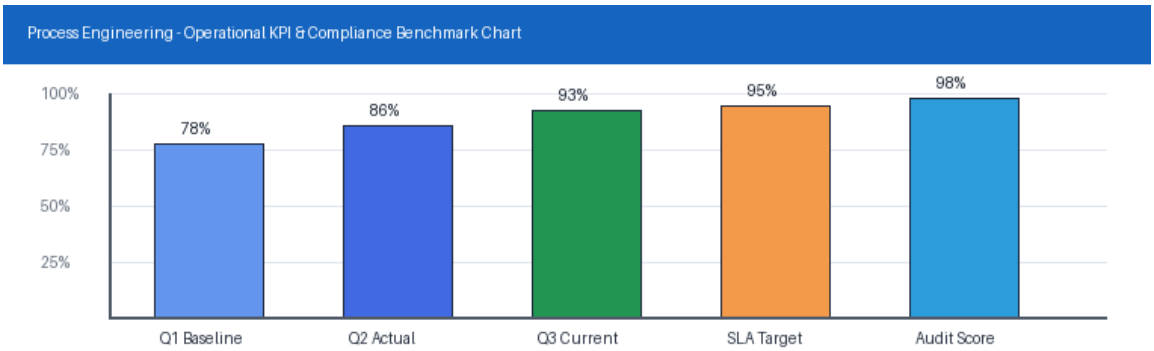
Line Balancing Study - Assembly Line 2 - Section 11: Continuous Improvement Roadmap

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 11 (Continuous Improvement Roadmap) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Schedules quarterly optimization milestones, automation targets, and capability reviews.

Operational requirement 11.1: Projected throughput improvement is approximately 6% without added headcount. Secondary control 11.2: Recommendation is scheduled for pilot validation next production cycle.

- Projected throughput improvement is approximately 6% without added headcount.
- Recommendation is scheduled for pilot validation next production cycle.
- Section 11 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-11	Carlos Ibarra	3 business days	Stage 11 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Line Balancing Study - Assembly Line 2 - Section 12: Appendix, Lineage & Sign-off

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 12 (Appendix, Lineage & Sign-off) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Records document lineage, governance metadata, version history, and owner sign-off.

Operational requirement 12.1: Recommendation is scheduled for pilot validation next production cycle. Secondary control 12.2: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.

- Recommendation is scheduled for pilot validation next production cycle.
- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Section 12 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-12	Carlos Ibarra	10 business days	Stage 12 Verified